Introducing IPUMS DHS-to-IHGIS Geographic Linkages

Attaching Contextual Data from Population and Agricultural Censuses to DHS Data

IPUMS Webinar, May 20, 2026

IPUMS IHGIS

IPUMS DHS

Agenda

- Intro to IPUMS DHS and IHGIS
- Overview of linkages
- Research example
- Linkage details
- Q & A

Intro to IPUMS DHS and IHGIS

IPUMS GLOBAL HEALTH	IPUMS IHGIS
Survey data	Census data
Microdata	Summary data
Displaced GPS points of DHS Clusters	GIS shapefiles (*Sub-state geographic detail*)

What is the DHS Program?

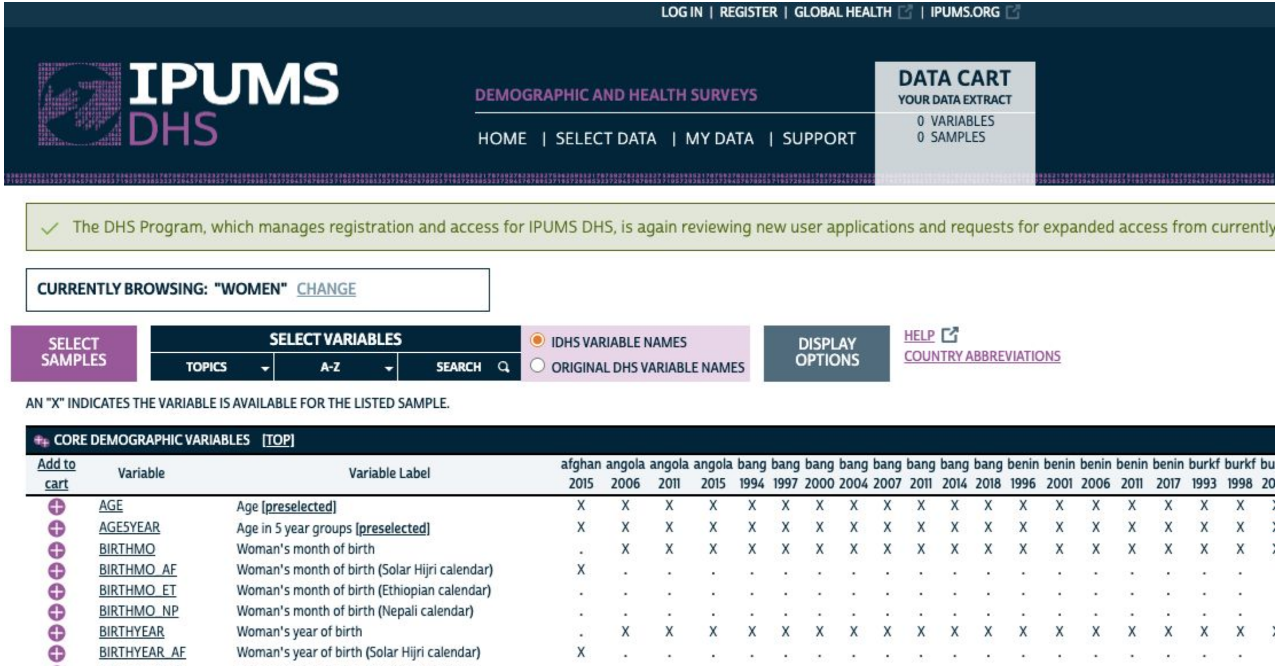
- Demographic and Health Surveys (DHS)
- 1980's to the present
- "Collect and disseminate accurate and representative data on population, health, HIV, and nutrition"
- Over 400 surveys in more than 90 countries
- Leading source of information on health in Low and Middle Income countries (LMICs)

https://dhsprogram.com/

What is IPUMS DHS?

Microdata from DHS public use files, recoded into over 20,000 harmonized variables

Additional features (file linkages, contextual variables, reproductive calendar)

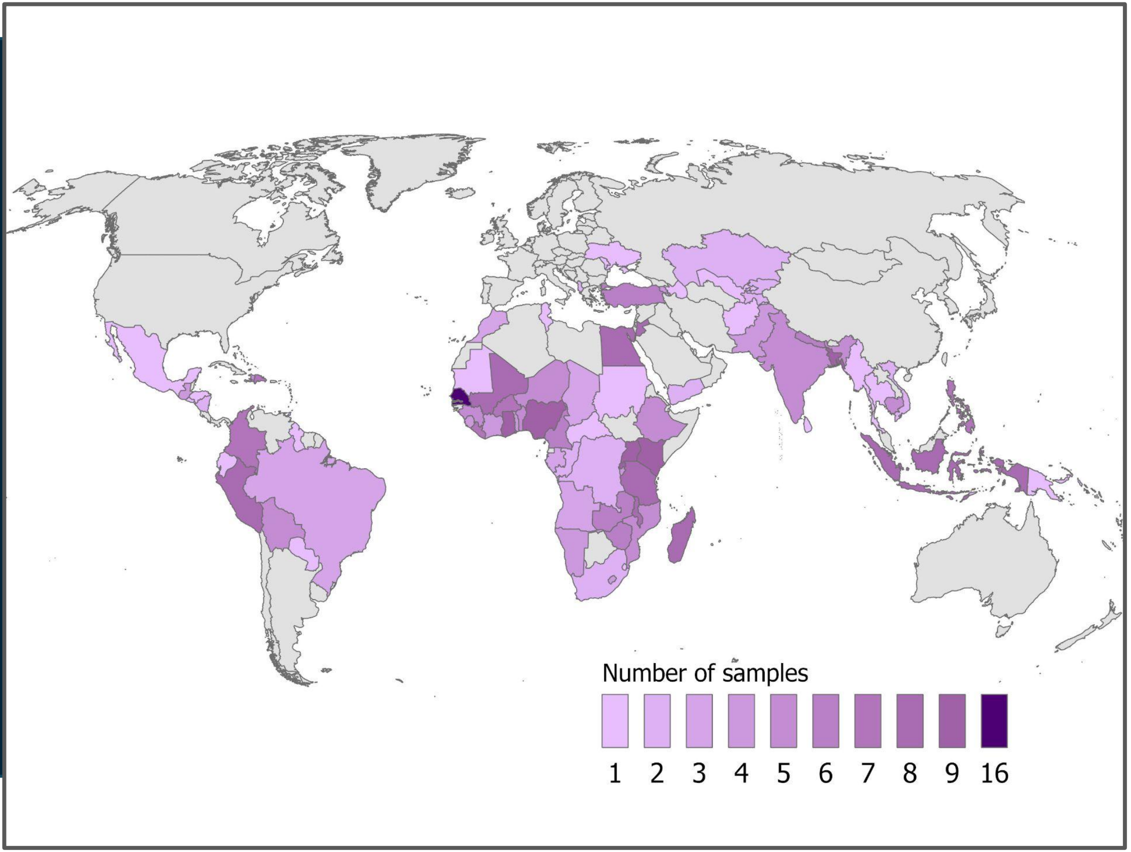


IPUMS DHS Variable Harmonization

INTEGRATED		Burkina Faso 2018	Kenya 2018	India 2018
100	**Never attended**	0 = Never	0 = Never	0 = Never
200	**Primary**	1 = Primary	1 = Primary	1 = Primary
300	**Post-primary/vocational**		6 = Post-primary/vocational	
400	**Secondary**			2 = Secondary
420	**Cycle 1**	2 = Secondary 1 cycle		
430	**Cycle 2**	3 = Secondary 2 cycle		
450	**A level**		7 = Secondary A level	
600	**Tertiary or higher**	4 = Tertiary		3 = Higher
610	**College**		5 = College	
620	**University**		4 = University	
650	**Postgraduate**			4 = Postgraduate

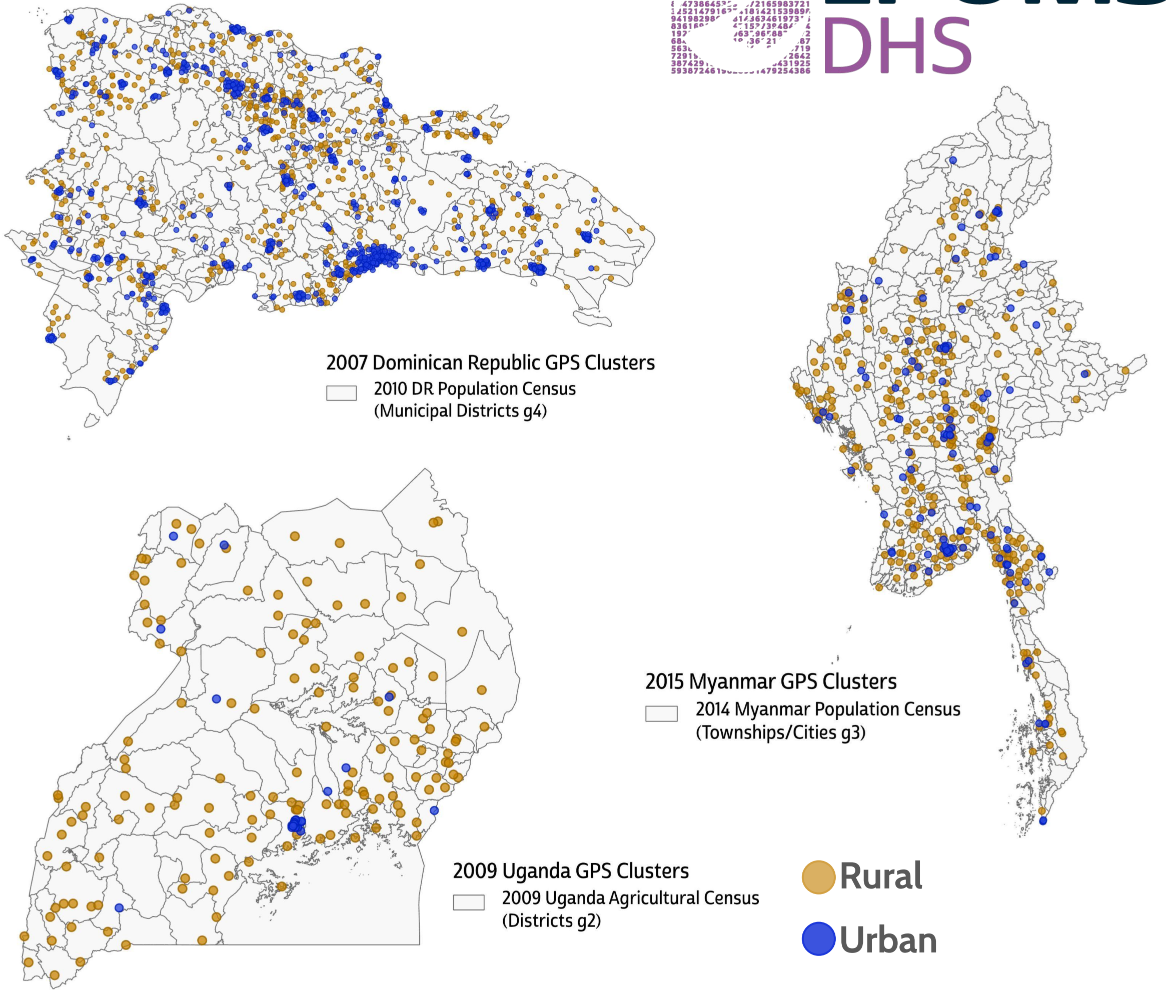
Samples and Countries in IPUMS DHS

- Nearly 350 samples
- 84 countries
- Low- and middle-income countries



DHS GPS Clusters

- DHS program provides randomly displaced GPS latitude/longitude georeferenced survey clusters (groupings of households in survey)
- Clusters are not displaced across survey regions or national boundaries.
- IPUMS DHS disseminates GPS clusters to authorized users



Contextual Variables in IPUMS DHS

Based on raster (gridded) data

Physical and Environmental

- Ecoregion
- Soil
- Normalized Difference Vegetation Index (NDVI)*
- Precipitation*
- Temperature (min, max)*

Economic and Social

- Livelihood index
- Population density*
- Malaria*

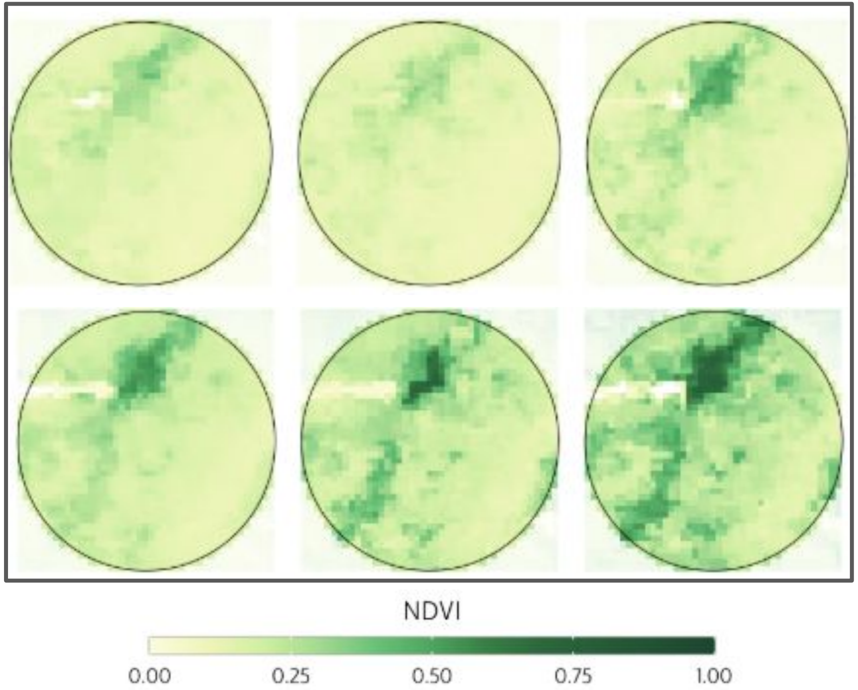
Agricultural

- Cropland
- Pastureland
- Crops harvested (17 major crops)
- Crop production (17 major crops)

* time variant variable

2003 NDVI Gridded Data

Aggregated to single DHS cluster point buffer (16-day intervals)



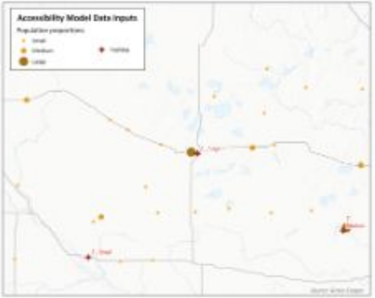
Spatial Analysis and Health Research Hub

https://tech.popdata.org/dhs-research-hub/

ACCESS TO HEALTHCARE PART 1: HOW IT IS MEASURED

HEALTH CARE | ACCESSIBILITY | DISTANCE MODELS | GRAVITY MODELS | POPULATION DENSITY | NETWORK ANALYSIS

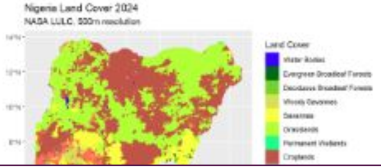
Many people in developing countries experience poor health as a result of poor spatial access to clinics and hospitals. Our understanding of how to improve global health is limited by our ability to measure healthcare accessibility.



VEGETATION AND LAND COVER PART 3: LAND COVER/LAND USE

LAND COVER | LAND USE CHANGE DYNAMICS | AGRICULTURE | TERRA | SF

An almost overwhelming number of vegetative and



CATEGORIES

- All (28)
- Accessibility (1)
- Agriculture (8)
- Cartography (3)
- CHIRPS (2)
- CHIRTS (4)
- Climate (2)
- Climatological normals (1)
- Comparable indicators (2)
- Comparison (1)
- Data sources (1)
- Distance models (1)
- Extreme weather (10)
- Fertility (1)
- Food production (4)
- Forecasts (2)

What is IHGIS?

IPUMS IHGIS

- Population and housing and agricultural census data
- Data from published tables
- GIS boundary files

Cuadro 2: PARAGUAY. PUEBLOS INDÍGENAS
Población por departamento, según familia lingüística y pueblo indígena, 2012

Familia lingüística	Pueblo Indígena	TOTAL	Concepción	San Pedro	Guairá	Caaguazú	Caazapá	Itapúa	Alto Paraná	Asunción y Central	Amambay	Canindeyú	Presidente Hayes	Boquerón	Alto Paraguay
	TOTAL	112.848	3.998	3.572	1.221	9.425	3.547	2.370	7.042	2.458	11.852	13.484	25.789	23.950	4.140
1. GUARANÍ	Guaraní Occidental	2.379	0	167	0										
	Aché	1.942	0	0	0										
	Ava Guaraní	17.697	142	1.524	0										
	Mbya	21.422	1.507	1.273	1.221										
	Paĩ Tavyterã	15.097	1.869	391	0										
	Guaraní Ñandéva	2.393	0	0	0										
2. LENGUA MASKOY	Toba Maskoy	2.817	0	0	0										
	Enlhet Norte	8.632	0	0	0										
	Enxet Sur	5.740	381	0	0										
	Sanapaná	2.833	0	0	0										
	Angaité	6.638	13	0	0										
	Guaná	86	86	0	0										
3. MATACO MATAGUAYO	Nivaclé	16.350	0	0	0										
	Maká	1.892	0	0	0										
	Manjui	385	0	0	0										
4. ZAMUCO	Ayoreo	2.481	0	0	0										
	Ybytoso	1.824	0	0	0										
	Tomárãho	183	0	0	0										
5. GUAICURU	Qom	2.057	0	217	0										

Fuente:
STP/DGEEC. III Censo Nacional de Población y Viviendas para Pueblos Indígenas 2012.

Table 1.8: Total Population by Broad Age Groups and Ethnicity, Zimbabwe 2012 Census

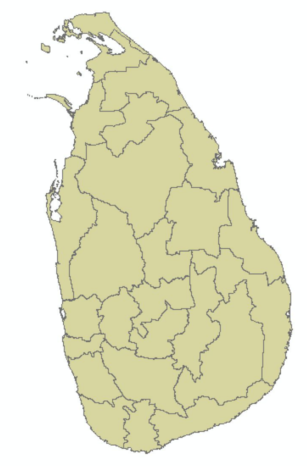
	Age Group					
Ethnic Origin	0 - 14	15 - 49	50-64	65+	Not Stated	Total
African	5351019	6239820	848450	522376	18355	12980020
European	4452	10994	6424	6521	341	28732
Asiatic	1857	5330	1865	977	126	10155
Mixed Race	5375	8745	2469	1300	34	17923
Other	222	524	197	134	7	1084
Not Stated	9356	4263	510	396	8800	23325
Total	5372281	6269676	859915	531704	27663	13061239

Table 1.9: Urban Population by Broad Age G Ethnicity, Zimbabwe 2012 Census

Ethnicity	0 - 14	15 - 49	50-64	65+
African	34.2	62.3	5.3	2.0
European	0.1	0.3	0.1	0.1
Asiatic	*	0.2	*	*
Mixed Race	0.1	0.2	0.1	*
Other	*	*	*	*
Not Stated	0.1	*	*	*
Total	34.5	63.0	5.5	2.2

Table 1.10: Rural Population by Broad Age G 2012 Census

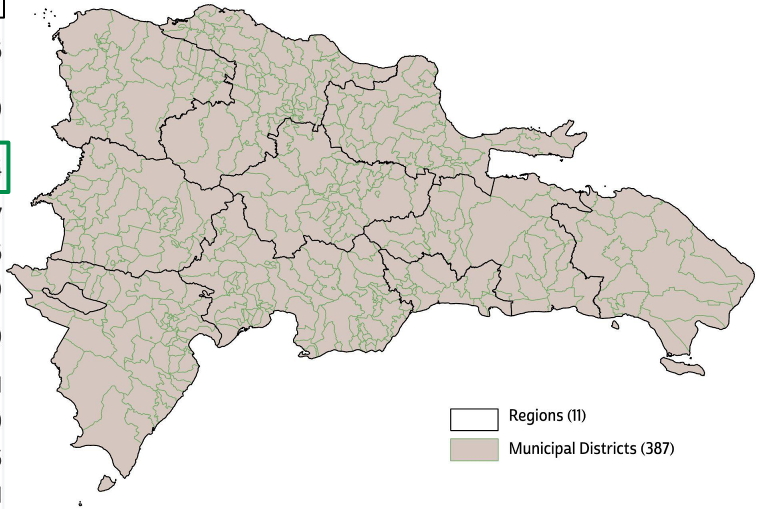
Ethnicity	0 - 14	15 - 49	50-64	65+
African	44.4	50.5	7.1	5
European	23	62.2	21.7	14.4
Asiatic	19.6	76.6	16.3	3.6
Mixed Race	40.9	52.9	11.2	6.1
Other	23.5	58.6	16.4	17.2
Not Stated	42.4	19.1	2.2	1.9
Total	44.4	50.5	7.1	5



Geographic Detail in IHGIS

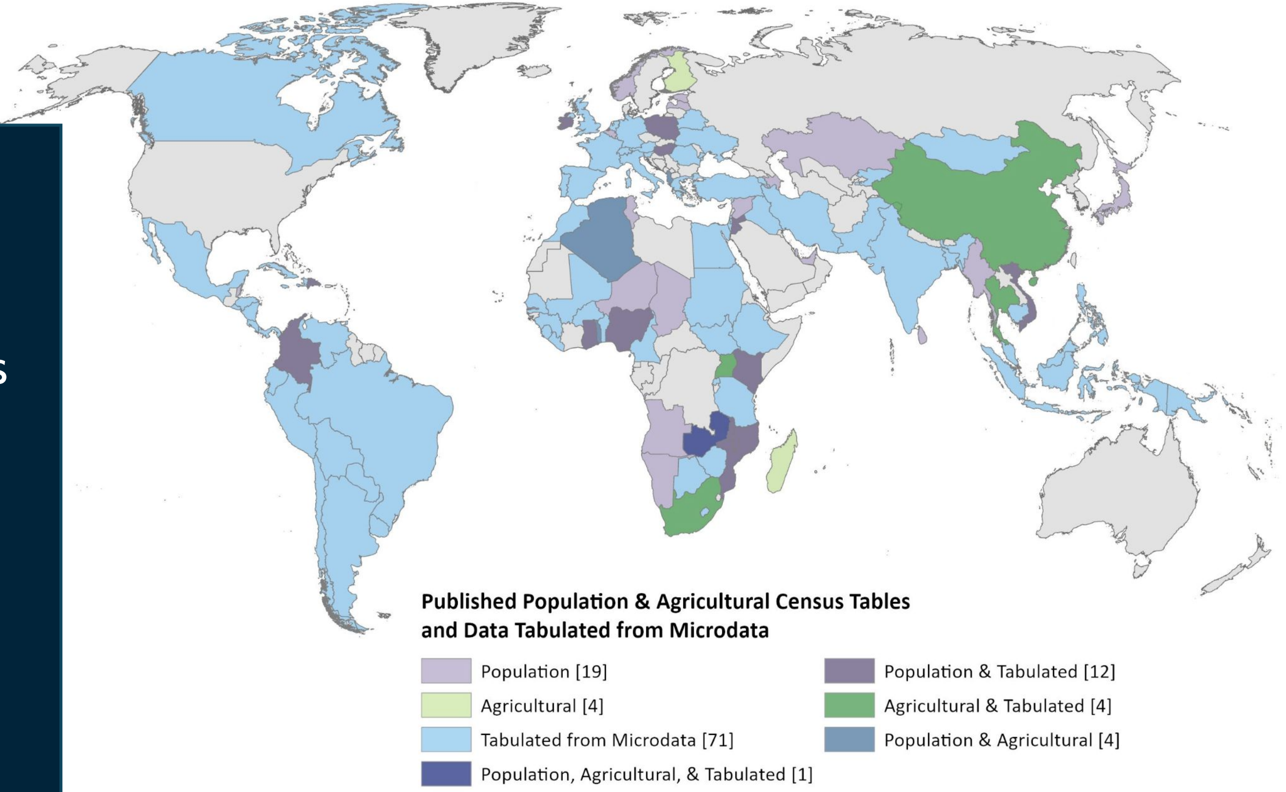
Dataset	Country	Year	Level	Tabulation Geography	# of Units	Mean Population	Mean Area	Data Tables
DO2010pop	Dominican Republic	2010	g1	Regions	11	858,662	4,374.6	55
DO2010pop	Dominican Republic	2010	g2	Provinces	33	286,221	1,458.2	56
DO2010pop	Dominican Republic	2010	g3	Municipalities	156	60,547	308.5	9
DO2010pop	Dominican Republic	2010	g4	Municipal districts	387	24,406	124.3	4
IE1991pop	Ireland	1991	g1	Provinces	4	881,430	17,563.0	17
IE1991pop	Ireland	1991	g2	County & County Borough groups	27	130,582	2,601.9	16
IE1991pop	Ireland	1991	g3	Counties	32	110,179	2,195.4	20
IE1991pop	Ireland	1991	g4	Rural districts/Urban districts/Municipal boroughs	217	16,248	323.7	19
IE1991pop	Ireland	1991	g5	Towns/Environs	254	7,645	0.0	1
KE2019pop	Kenya	2019	g1	Counties	47	1,012,006	12,363.0	49
KE2019pop	Kenya	2019	g2	Sub-Counties	349	136,287	1,664.9	35
KE2019pop	Kenya	2019	g3	Divisions	1,000	47,564	581.1	1
KE2019pop	Kenya	2019	g4	Locations	3,838	12,393	0.0	1
KE2019pop	Kenya	2019	g5	Sub-Locations	8,932	5,325	0.0	2

Dominican Republic 2010 Population Census Tabulation Geographies



IHGIS Datasets

- Published data from 45 countries
  - 43 Population & Housing censuses
  - 15 Agricultural censuses
- 305 tabulated microdata datasets
- 1966-2021



Overview of linkages

Linking IPUMS DHS and IHGIS

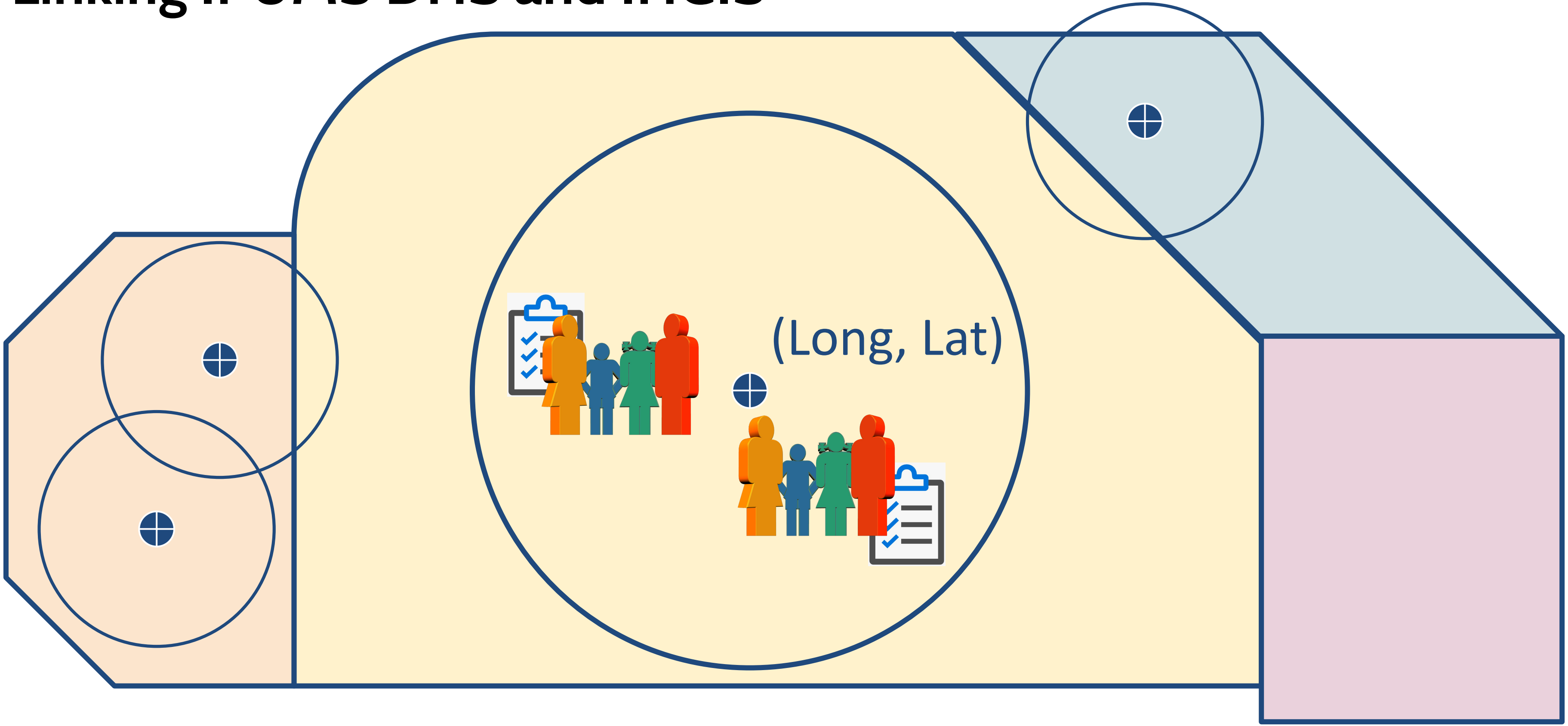
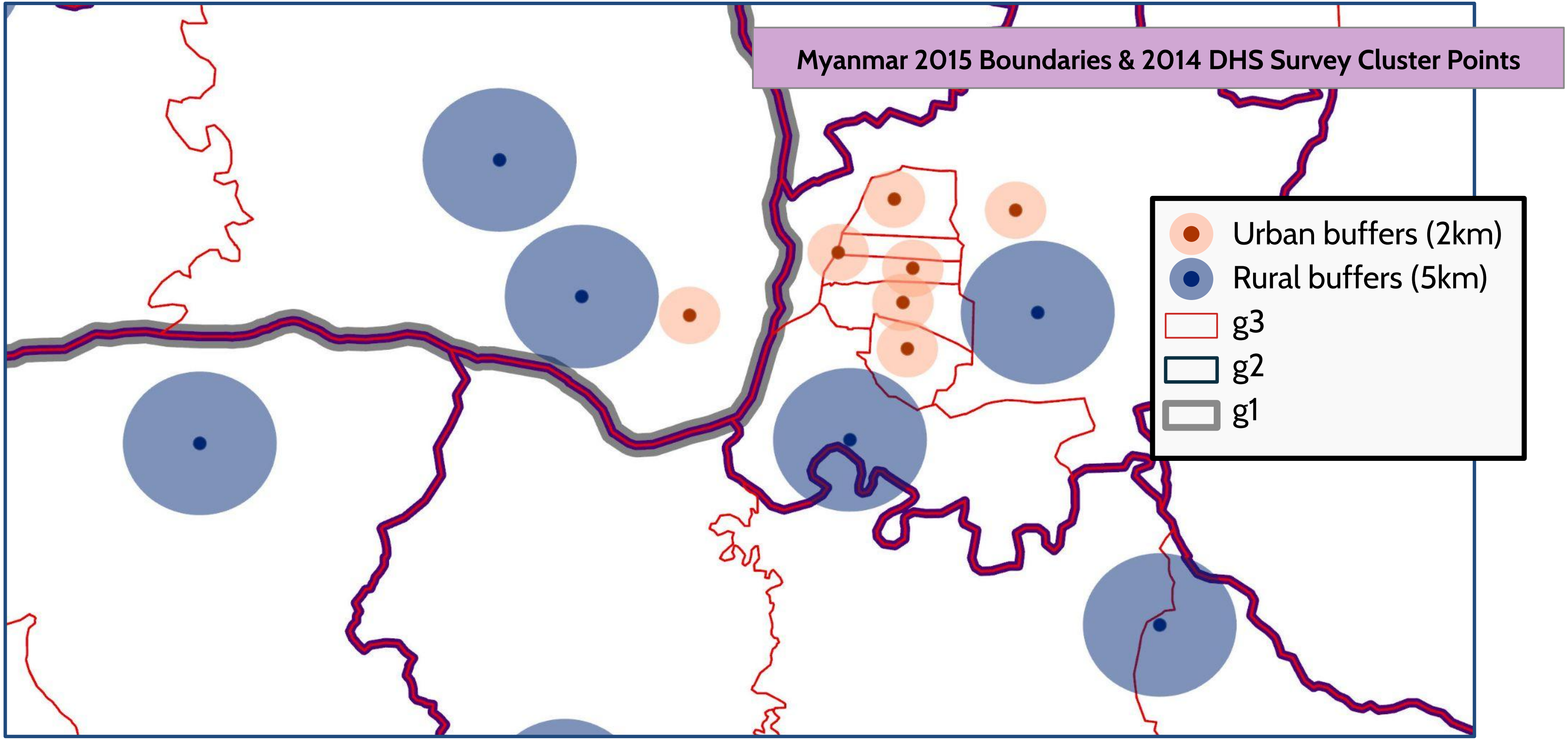


Image of family by Gerd Altmann from Pixabay

Census-to-DHS Geography

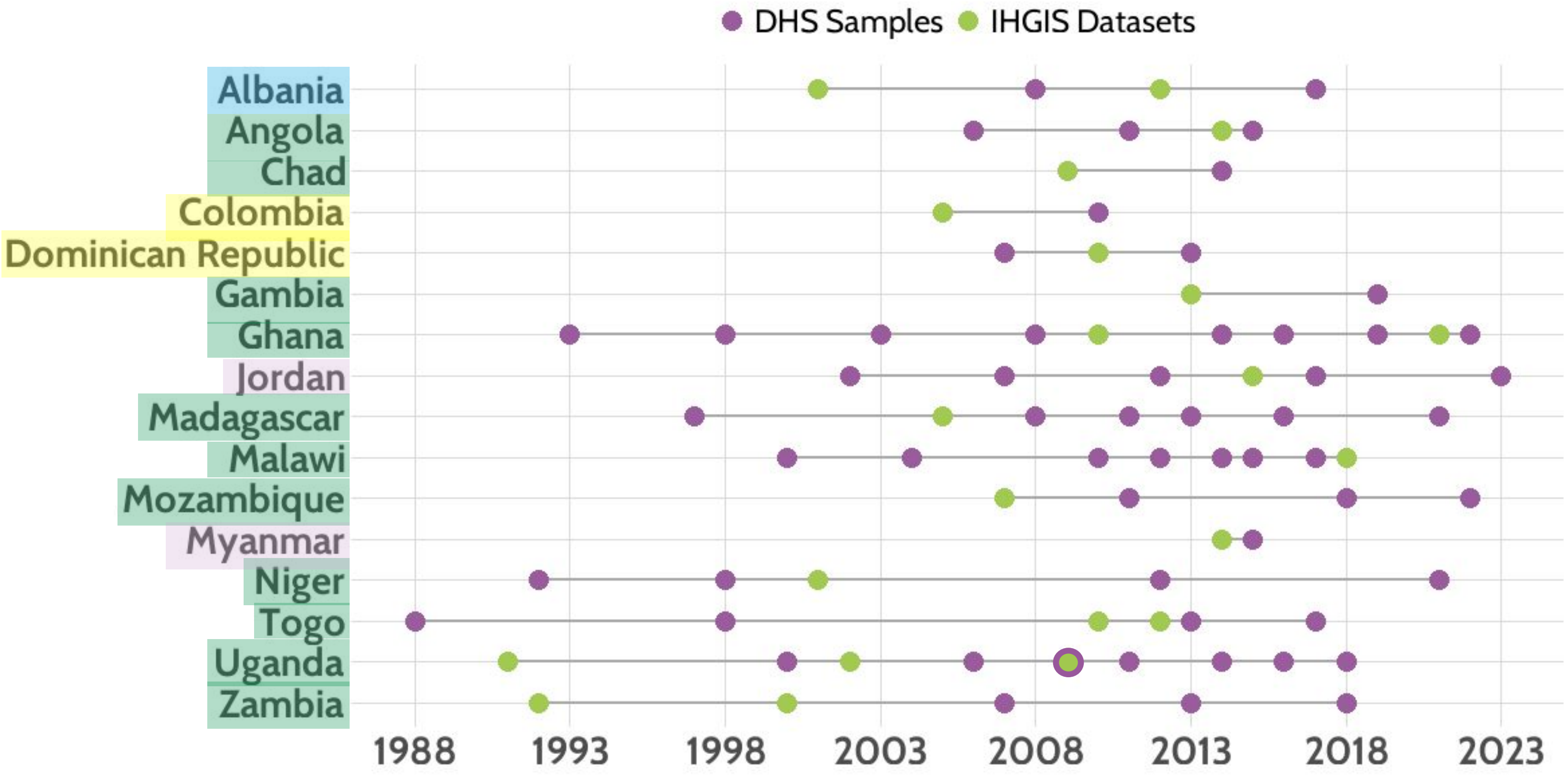


IPUMS DHS-IHGIS Linking Variables

DHS records with cluster ID, linking variable, and probability

DHSID	IHGISUG2009AGG2	IHGISUG2009AGG2P	Age
UG200900000020	UG104	0.753	42
UG200900000023	UG108	1	34
UG200900000024	UG108	0.922	44
UG200900000025	UG110	1	33
UG200900000025	UG110	1	37
UG200900000026	UG110	1	40
UG200900000027	UG110	1	42

Available IHGIS-DHS Links by Country and Year



Research example

Example Research Question

What is the relationship between child stunting and district-level food shortage in Uganda?

Image: WHO

Image: allAfrica

Available Data

IPUMS DHS Uganda 2011 Sample

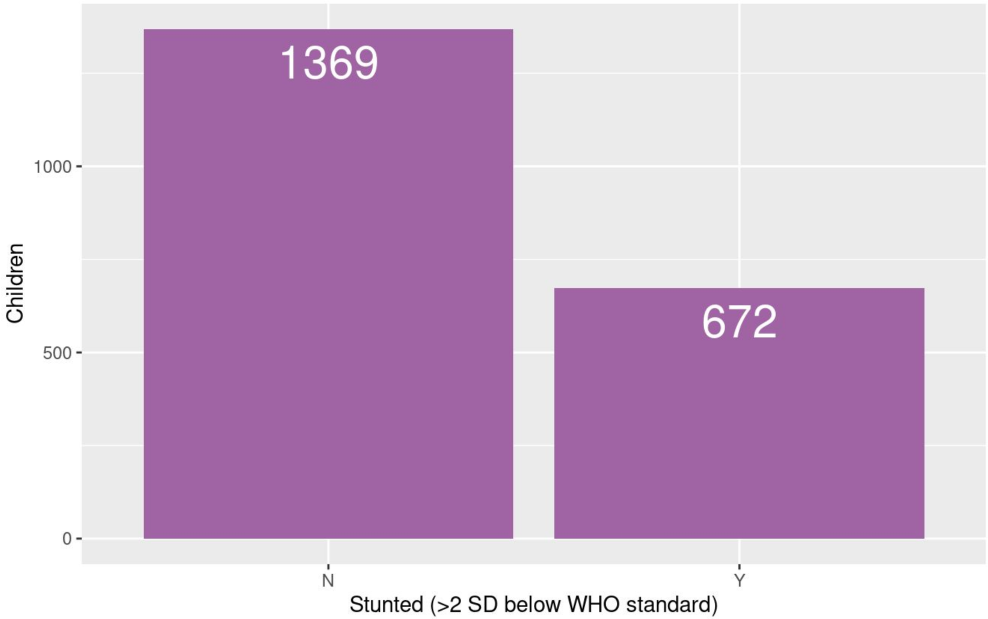
- Child records
- Height-for-age z-score as measure of stunting (HWHAZWHO)
- Child- and household-level controls
- Linking variables

IHGIS Uganda Census of Agriculture, 2008-09

- Table: Agricultural Households that Experienced Food Shortage (AAX)
- District records (n = 80)

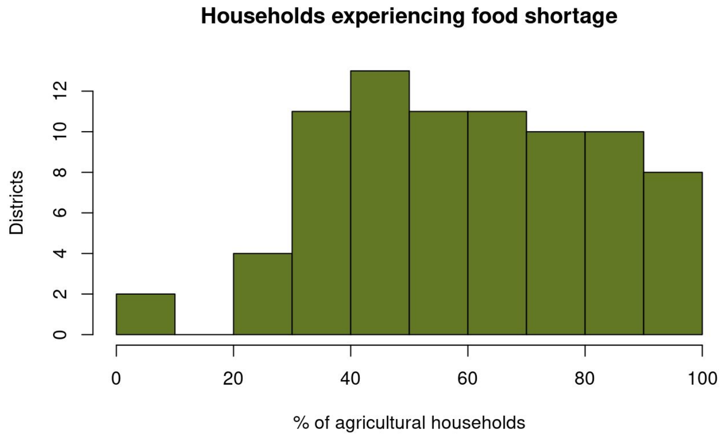
Live Demo Downloading IPUMS DHS & IHGIS Data

Data Exploration - Stunting



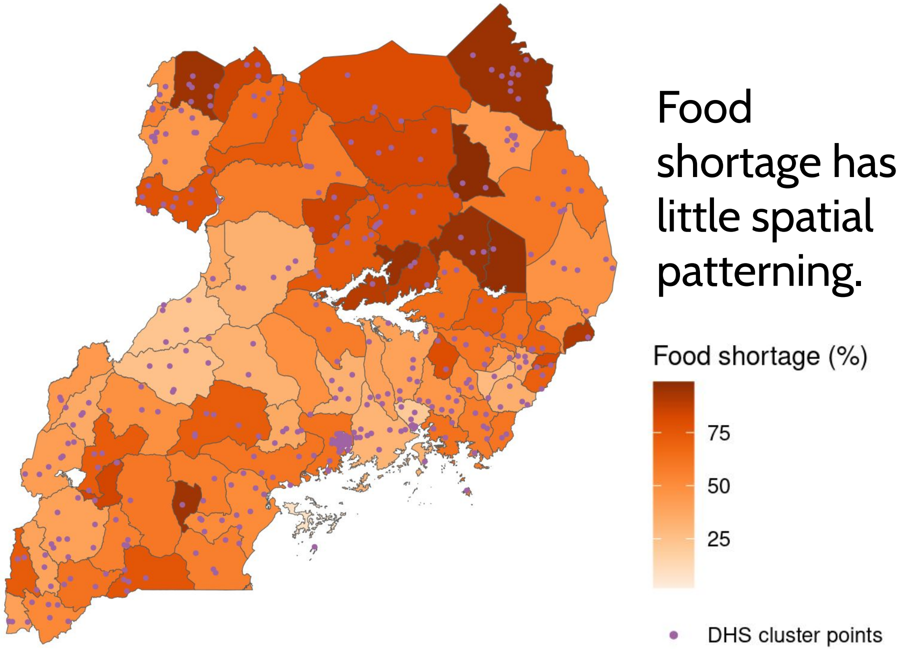
Nationwide, **33%** of children measured meet the WHO definition of stunting.

Data Exploration - Food Shortage



Between 30 and 100% of households experiencing food shortage, the number of districts is roughly uniform.

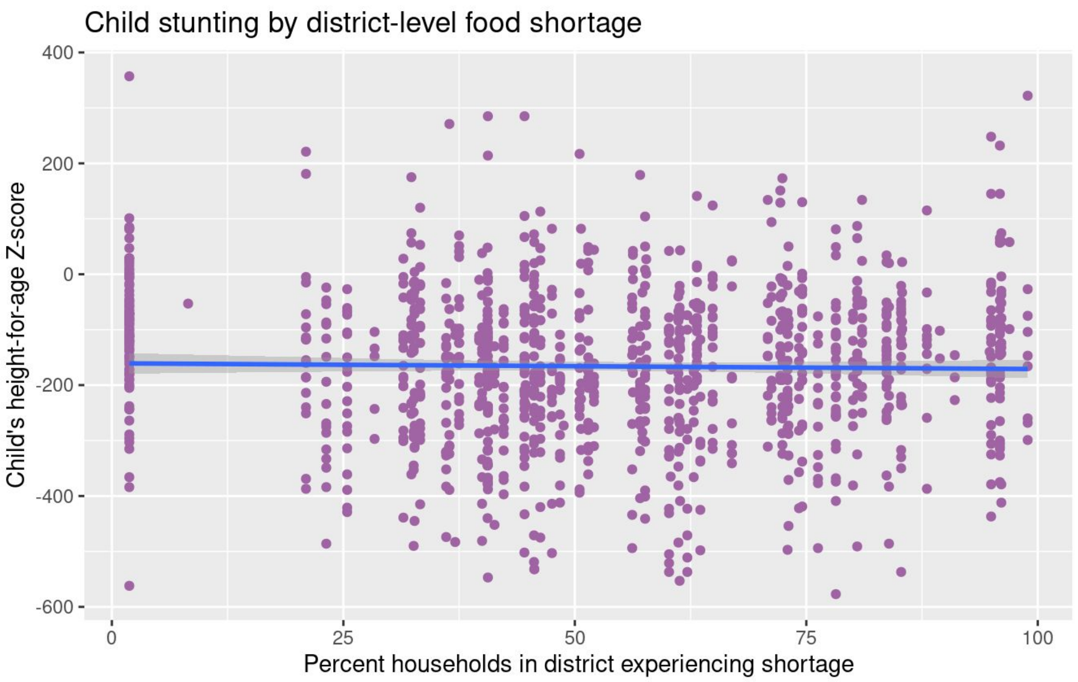
% Agricultural households experiencing food shortage



Food shortage has little spatial patterning.

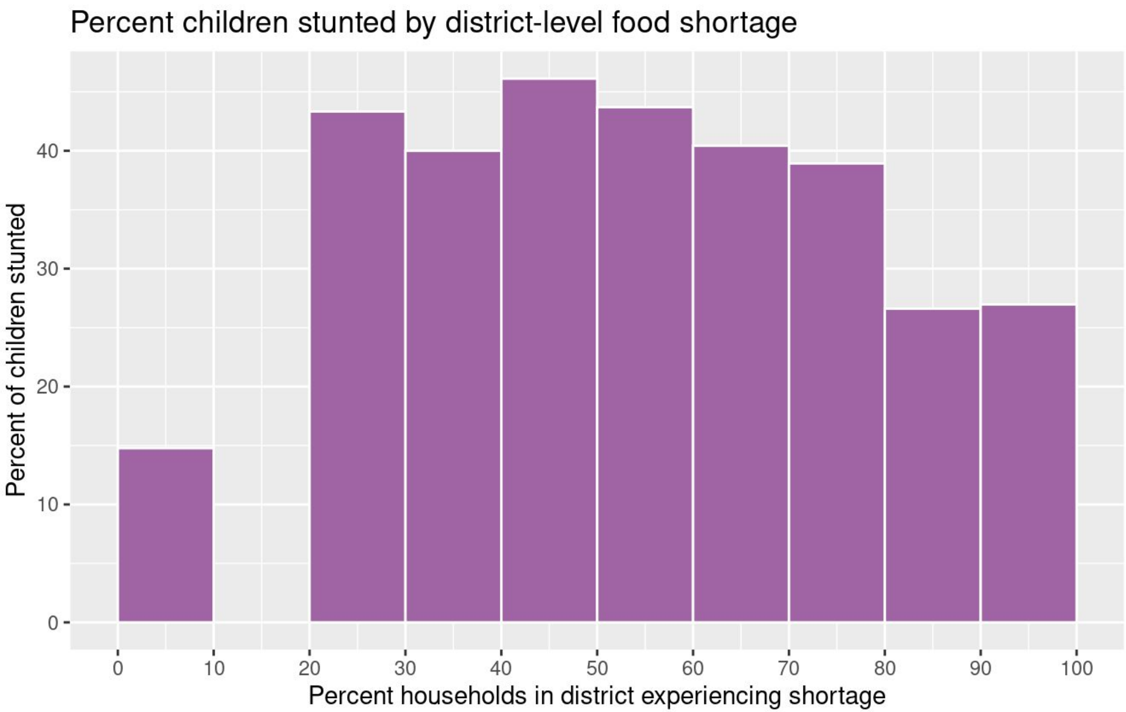
Data Exploration - Stunting & Food Shortage Scatter Plot

Scatter plot shows little relationship, except that all children in a district are at the same value of percentage of households experiencing food shortage.



Data Exploration - Stunting & Food Shortage Bar Chart

Interestingly, as the percent of households in a district experiencing food shortage increases above 50%, the percent of children stunted decreases.



Stunting & Food Shortage Logit Model

stuntWHO ~ AGE + EDYRTOTAL + UNIMPTOILETHH + KIDAGEMO + as.factor(URBAN) + pct_shortage

data = filter(DHS11_IHGIS09, stuntWHO != “NIU” & KIDAGEMO > 23)

Controlling for

- Mother’s age
- Mother’s years of education
- Unimproved toilet
- Child’s age
- Household in rural area

Sample limited to children

- For whom stunting was measured
- More than 23 months old, who would have been conceived during or before food shortage reference period

Stunting & Food Shortage Logit Model Results

An increase of 1% of households in a district experiencing food shortage in 2009 is associated with a 1.23% decrease in the probability of a child being stunted in 2011.

Coefficient	Estimate	z value	Pr(>|z|)	
(Intercept)	1.47	3.13	0.00185	**
Mother's age	-0.0397	-3.80	0.000142	***
Mother's yrs. of education	-0.743	-3.84	0.000121	***
Household uses unimproved toilet	0.234	1.77	0.0769	.
Child's age (mo.)	-0.0267	-4.04	5.25e-05	***
Rural area	1.41	6.39	1.64e-10	***
% of households w/food shortage	-0.0124	-4.11	3.94e-05	***

Additional Research Ideas

- How do types of crops grown affect…
  - children's enrollment in school? *Some crops are harvested by children.*
  - prevalence of breastfeeding through agricultural labor demands on women
  - dietary diversity
- How are children's vaccination rates related to community-level education?
- How is community water supply and waste disposal infrastructure related to the incidence of diarrhea?

Linkage details

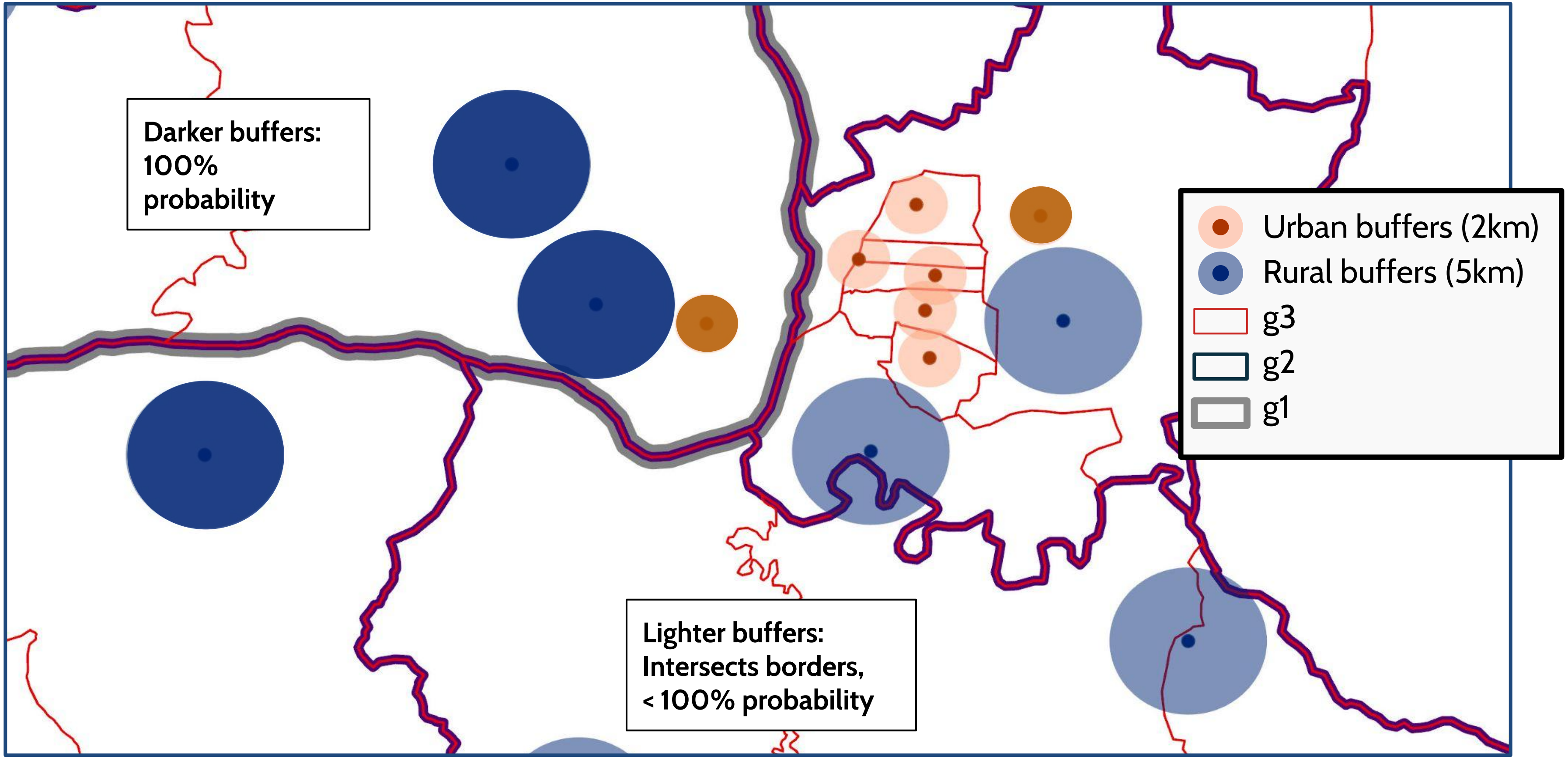
DHS-IHGIS Crosswalk Files

One file per
DHS sample - IHGIS tabulation geography

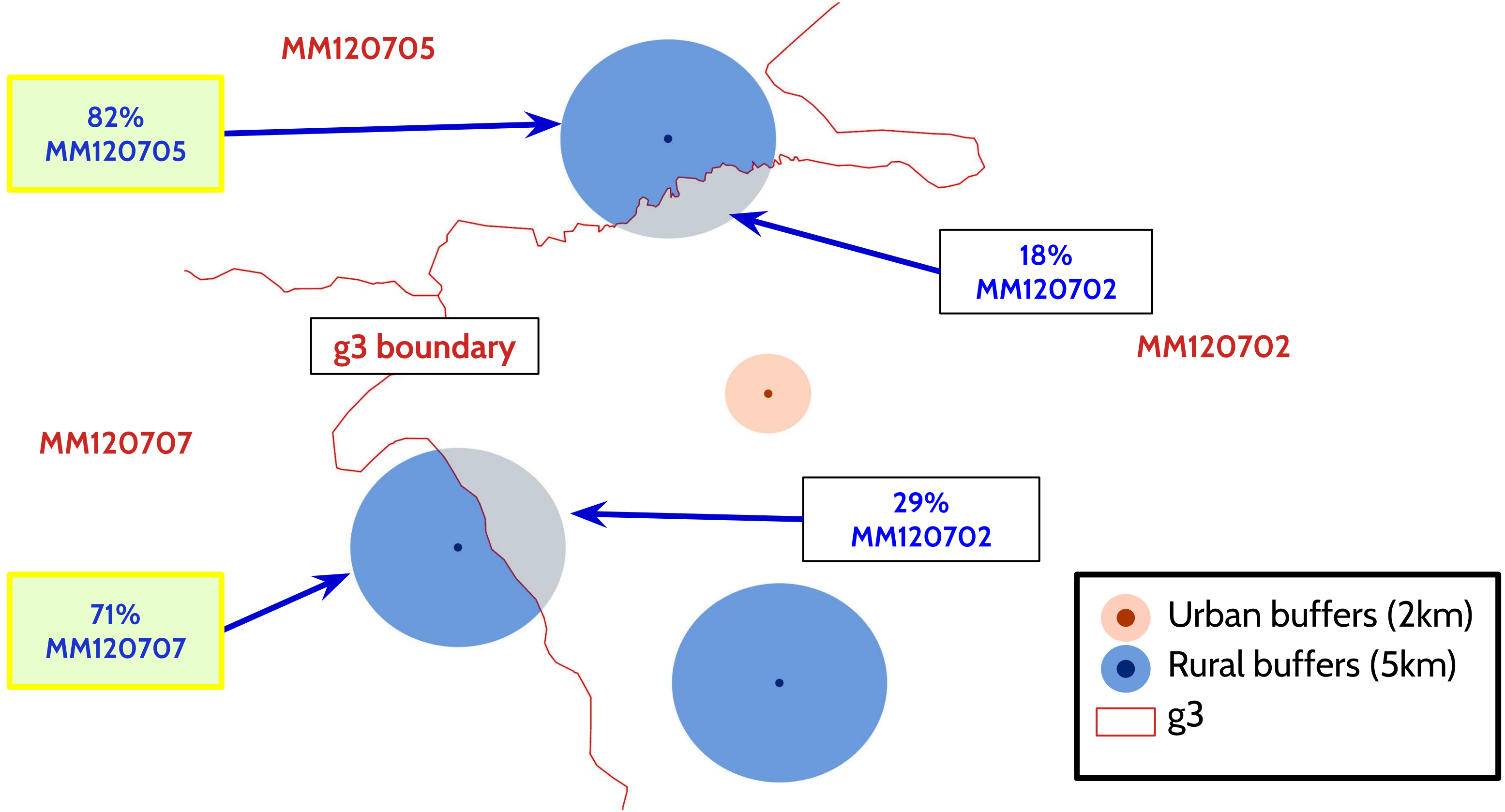
- DHSID - DHS cluster point identifier
- DHSYEAR - DHS sample year
- IHGIS_GISJOIN - IHGIS geographic unit identifier
- IHGIS_NAME - IHGIS geographic unit name
- IHGIS_PROP - Probability that this cluster point is in this IHGIS unit

DHSID	DHSYEAR	IHGIS_GISJOIN	IHGIS_NAME	IHGIS_PROP
UG201100000100	2011	UG109	Mpigi	1
UG201100000101	2011	UG109	Mpigi	0.873
UG201100000101	2011	UG107	Masaka	0.127
UG201100000103	2011	UG109	Mpigi	1
UG201100000105	2011	UG109	Mpigi	1
UG201100000107	2011	UG109	Mpigi	0.874
UG201100000107	2011	UG116	Wakiso	0.126
UG201100000109	2011	UG109	Mpigi	1
UG201100000110	2011	UG106	Lyantonde	1
UG201100000111	2011	UG114	Rakai	1
UG201100000120	2011	UG114	Rakai	0.746
UG201100000120	2011	UG107	Masaka	0.254
UG201100000122	2011	UG115	Ssembabule	1

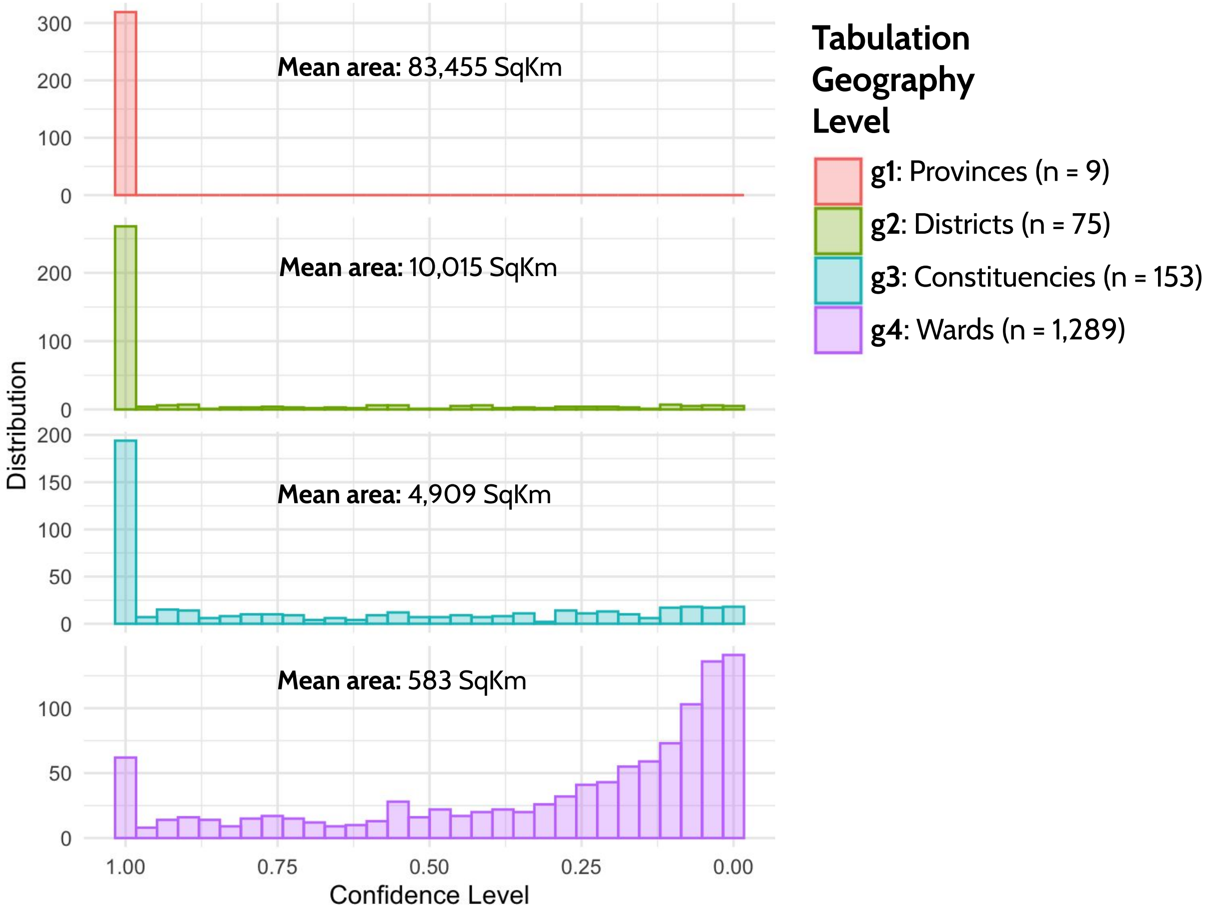
Computation of IHGIS-to-DHS Geography Crosswalks



Uncertainties in Unit Placement

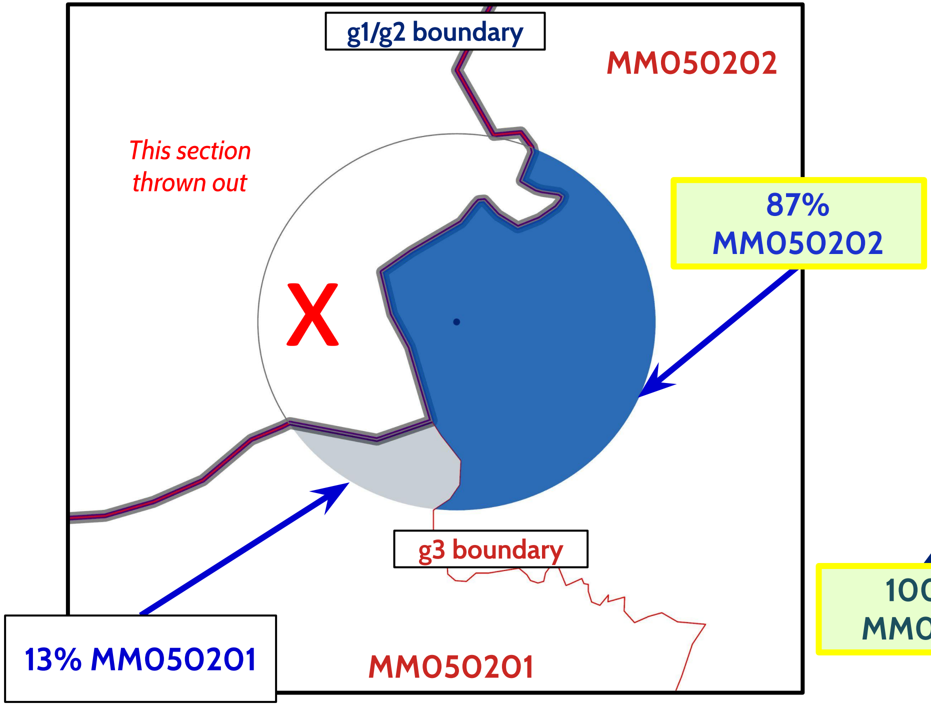


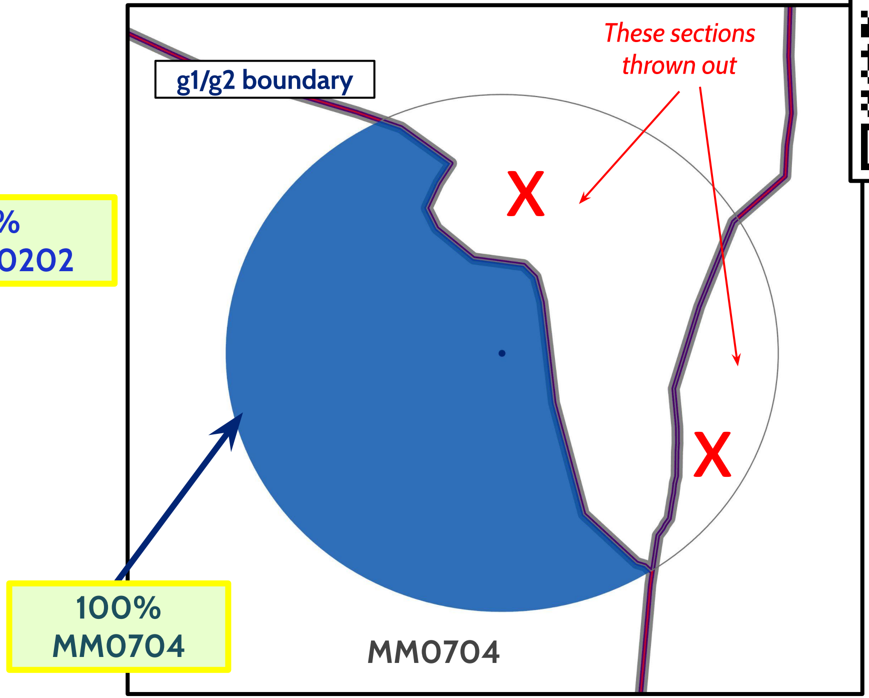
Placement confidence decreases as IHGIS units get smaller



Zambia
2007 DHS Sample GPS Locations in 2000 Population Census Tabulation Geographies

Container Units





For more details on DHS displacement procedures

IPUMS IHGIS

IPUMS DHS

https://ihgis.ipums.org

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https://dhs.ipums.org

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Hands-On Exercise

Github Repo Link: https://z.umn.edu/b7k1